

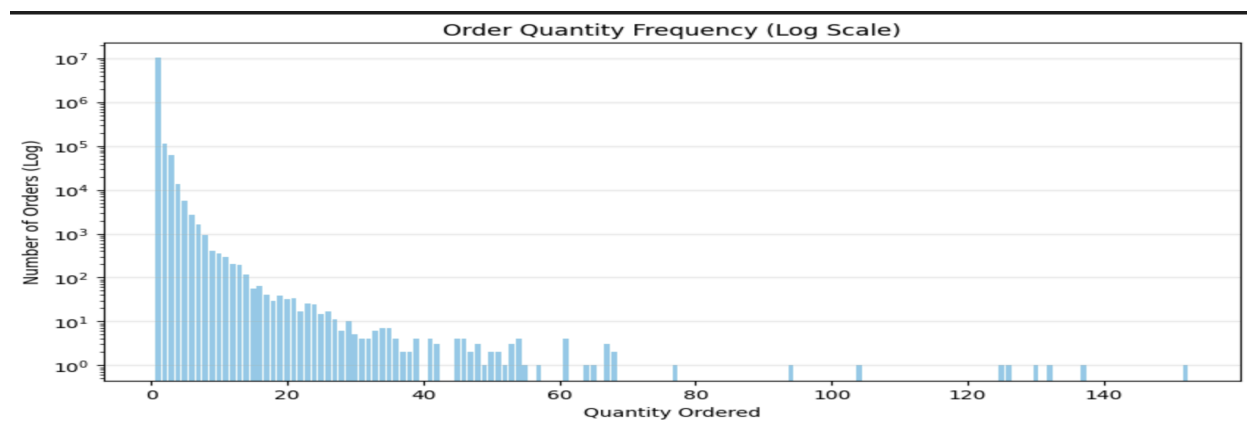
Part 1 - Data Understanding

1. How would you structure the data for forecasting?

The raw dataset contains order-level transactions that need to be transformed into forecastable time series before any modeling can happen. This requires a sequence of decisions across population separation, aggregation, and feature engineering.

Population separation

The first step is identifying and separating two structurally distinct purchasing populations. Going over the distribution of order quantity, I noticed some customer_ids fell under large bulk orders. This leads to the following assumption: If a customer_id bulk orders repeatedly, it's probably not a regular customer, and should be flagged as an indicator for the model to leverage. Encoding these patterns might yield more accurate forecasts, especially if the bulk-order behavior follows a pattern.



```
> ✓ 9.0s
print(order_df[order_df.is_bulk_buyer==True].revenue.mean())
print(order_df[order_df.is_bulk_buyer==False].revenue.mean())

13002.989621099166
1966.2278472433297
```

Repeat bulk buyers are classified as business accounts using a customer-level rule: more than 2 orders with quantity above 10, representing over 50% of their total order history. This yields 11 business accounts out of 10M+ rows. Rather than modeling them separately (too sparse for a standalone model), bulk presence is retained as a covariate: bulk order count per period. Both populations contribute to the OTB revenue target.

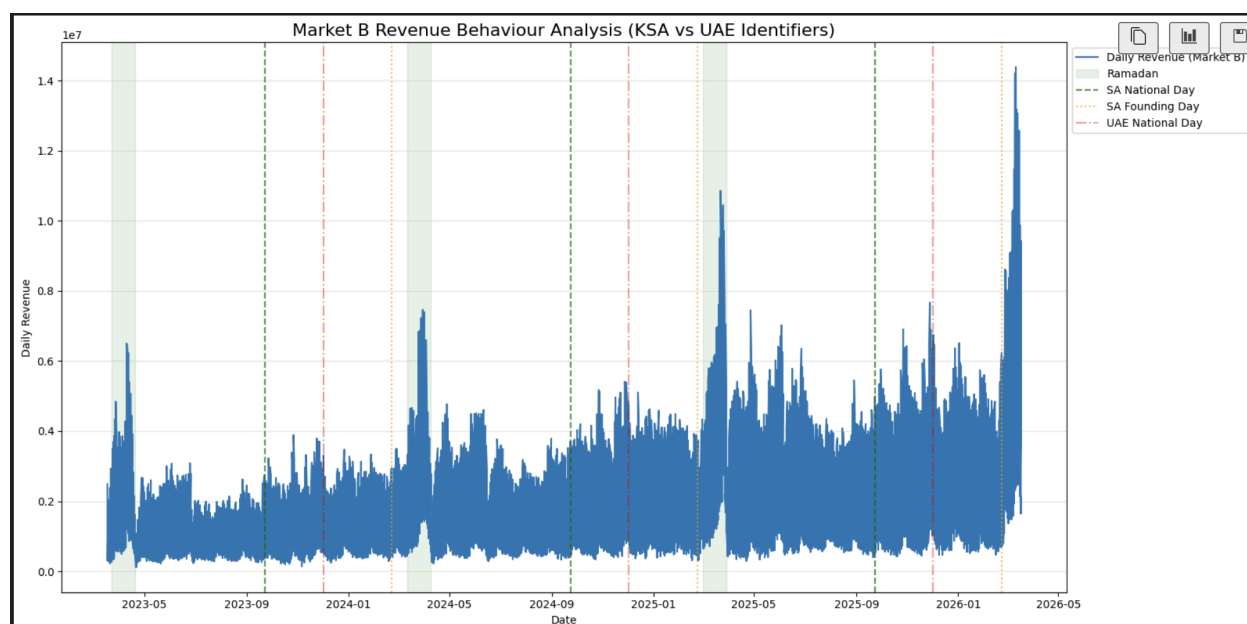
Cancellation treatment

18% of rows carry zero revenue despite a recorded order quantity. With only 1 negative revenue row in the entire dataset, order returns are not encoded here. Zero revenue rows are treated as cancelled orders, flagged as `is_cancelled`. Cancellation count is retained as a per-period covariate because as we forecast revenue, category x market pairs that lead to frequent returns are likely to serve as a signal.

Aggregation

Consumer and bulk orders are aggregated daily at date x category x market granularity. Each period produces: net revenue (sum), total order quantity (sum), bulk order count, and cancellation count. The dataset covers 5 categories across 2 markets (UAE and KSA), yielding 10 series.

While it was not explicitly specified that Market A/B correspond to UAE and KSA respectively, I ran a visual check by analyzing market behaviour on National holidays. Market B reacted predictably to Saudi Holidays. Holiday markets will be further explained in the following section below. Here's the visual I used for confirmation:



I ran the check on both A/B and found that patterns aligned.

Feature engineering

All calendar and event features are engineered post-aggregation on the resulting time series.

The GCC context drives the feature set:

Ramadan is encoded as an exponential weight from 0 to 1 across the month, capturing the known behavioral pattern of low activity early in the period accelerating sharply toward Eid al-Fitr. This behavioural pattern is based on the following assumption: Ramadan is a spiritual month where in the early days consumption across the board falls, but as Eid gets closer,

consumer behaviour shifts to gift buying and Eid shopping. Instead of using a simple binary 0/1 tag, I hypothesized that the shift is actually gradual and follows an exponential distribution. Therefore, a normalized weight from an exponential function is used to model this behaviour.

Dates are hardcoded per year from official UAE moon sighting announcements rather than computed algorithmically. Eid al-Fitr and Eid al-Adha are flagged as 3-day binary events.

```
# Updated to match official UAE Moon Sighting Committee announcements
RAMADAN_PERIODS = {
    2023: ("2023-03-23", "2023-04-20"),
    2024: ("2024-03-11", "2024-04-09"),
    2025: ("2025-03-01", "2025-03-29"), # Actual UAE end date
    2026: ("2026-02-18", "2026-03-18"), # Current official UAE end date
}

EID_AL_FITR = {
    2023: ["2023-04-21", "2023-04-22", "2023-04-23"],
    2024: ["2024-04-10", "2024-04-11", "2024-04-12"],
    2025: ["2025-03-30", "2025-03-31", "2025-04-01"], # Actual UAE dates
    2026: ["2026-03-19", "2026-03-20", "2026-03-21"], # Official UAE dates
}

EID_AL_ADHA = {
    2023: ["2023-06-28", "2023-06-29", "2023-06-30"],
    2024: ["2024-06-16", "2024-06-17", "2024-06-18"],
    2025: ["2025-06-06", "2025-06-07", "2025-06-08"],
    2026: ["2026-05-27", "2026-05-28", "2026-05-29"], # Current projection
}
```

Market-specific holidays are encoded separately: UAE National Day (Dec 2) and DSF window (Jan 15 to Feb 15) for Market A, Saudi National Day (Sep 23) and Founding Day (Feb 22) for Market B.

New Year and the Christmas window are included to capture the significant expat population effect in UAE. Standard calendar features cover week of year, month, quarter, and weekend (Friday and Saturday in GCC).

All features are forward-generatable from known dates, meaning the full feature set for the 12-month forecast horizon can be constructed.

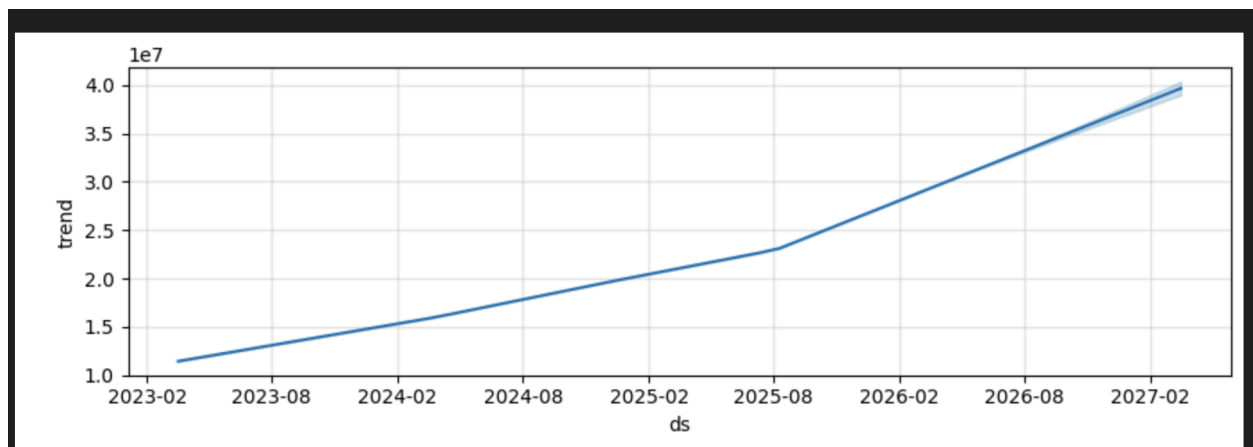
2. Seasonality Analysis

Prophet decomposition was run on three series: overall, Market A (UAE), and Market B (KSA).

The decomposition separates trend, weekly seasonality, and yearly seasonality.

Trend

The overall series shows strong linear growth from approximately 12M to 40M in daily revenue over the 3-year history.

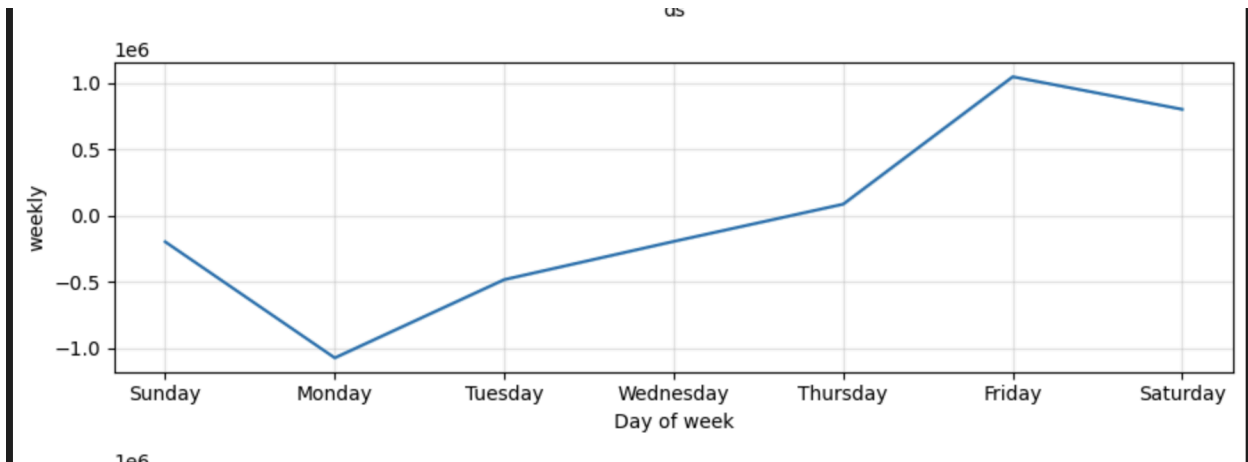


Weekly seasonality

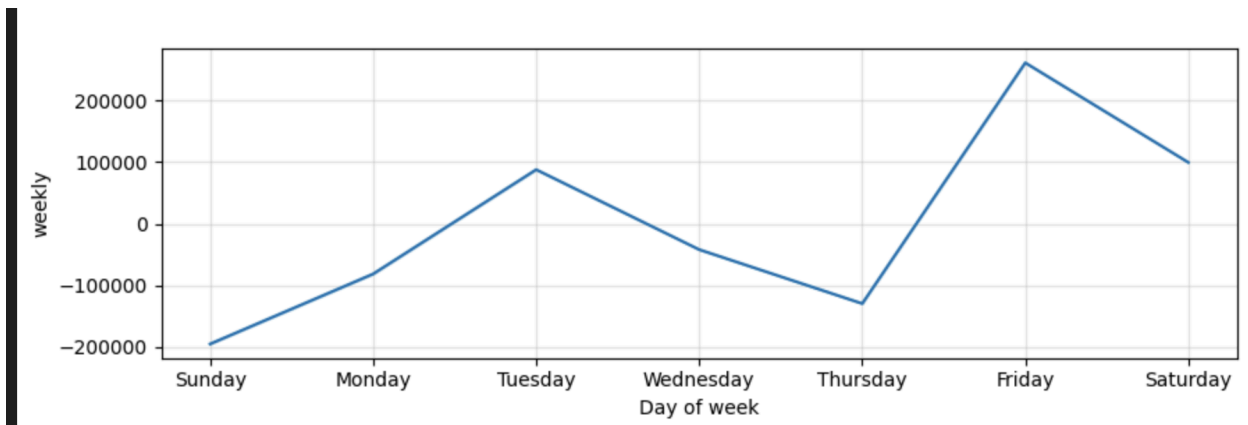
Friday is the dominant peak across both markets, consistent with GCC weekend shopping behavior. Monday is the trough in Market A. Market B shows a different weekly rhythm, with

Sunday as the trough and an unexpected Tuesday secondary peak not present in Market A. This difference is structurally meaningful and justifies exploring modeling the two markets independently.

Market A:



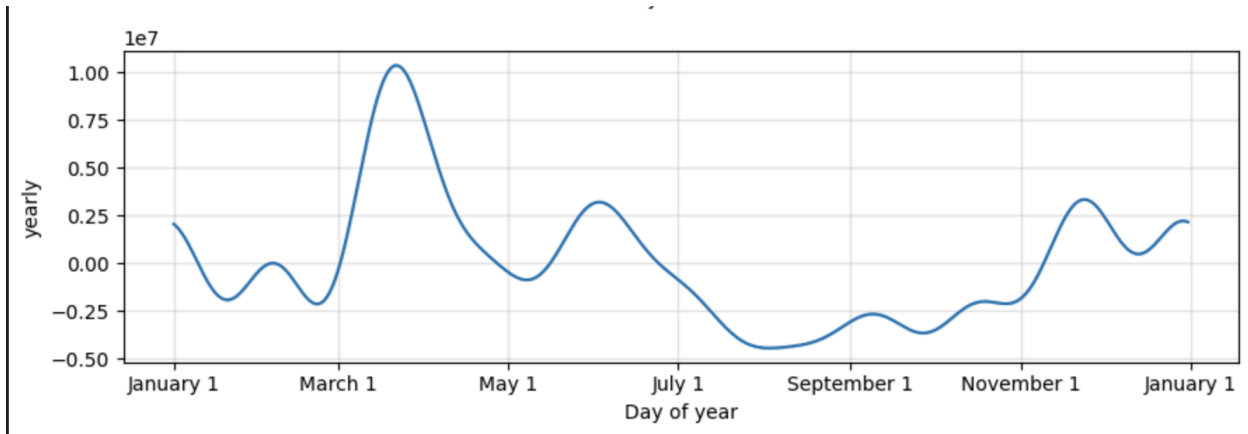
Market B:



Yearly seasonality

The March peak is the dominant annual signal across both markets, corresponding to Ramadan and Eid al-Fitr. The relative magnitude differs: in Market B (KSA), the Ramadan peak is proportionally larger and the rest of the year sits closer to or below baseline, reflecting a more

concentrated seasonal demand pattern. In Market A (UAE), the seasonal profile is more distributed, with secondary peaks in May to June (Eid al-Adha), and a November to December recovery aligning with the Dubai Shopping Festival window and year-end activity. The summer trough (July to September) is present in both markets, consistent with the GCC heat and travel season.



3. Dataset Challenges

Returns not captured. Only 1 negative revenue row exists in 10M+ records. Return activity is a real demand signal in luxury e-commerce and its absence is a data architecture limitation. A production system would incorporate a dedicated returns feed.

Cancellation volume. 18% of rows carry zero revenue. These are treated as cancellations, but the dataset provides no explicit cancellation flag to validate this assumption. If some zero-revenue rows represent other transaction types (full discounts, data issues), the cancellation covariate will carry noise.

Bulk and consumer mixing. Without explicit account type labeling in the source data, business account classification relies on behavioral inference. The rule (more than 2 orders above quantity 10, over 50% of order history) is defensible but imperfect. Edge cases exist where high-frequency consumers could be misclassified.

Islamic calendar shift. Ramadan moves approximately 11 days earlier each year on the Gregorian calendar. With 3 years of history, the model has seen Ramadan in March, then February, with limited exposure to how demand behaves when Ramadan falls in different seasonal contexts.